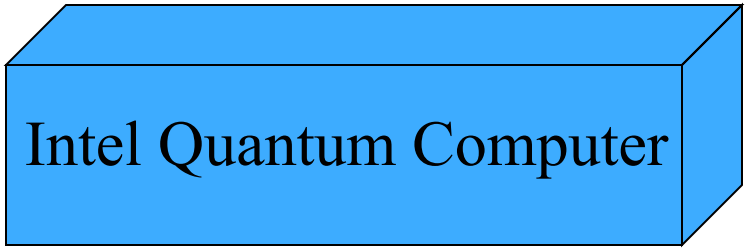
Quantum Error Correction

Weatherproofing Quantum Computers





Dave Bacon

Institute for Quantum Information

California Institute of Technology

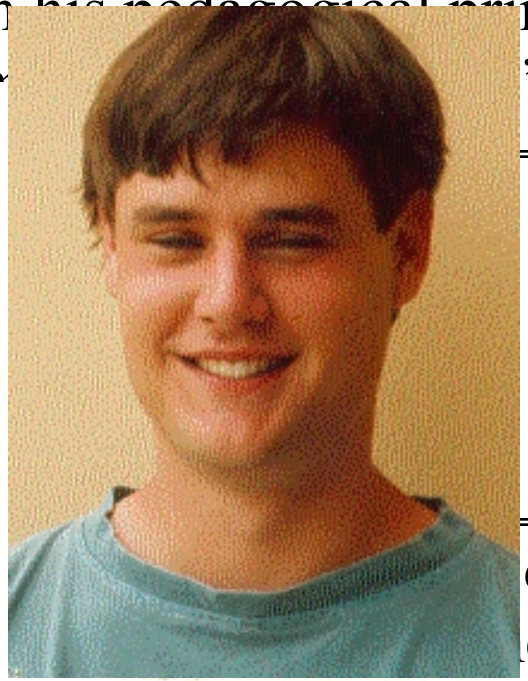
“‘My old school master was a demigod, an iron man, and woe be unto that boy who could not answer yes or no to a direct question,’ writes Kierkegaard about Michael Nielsen.

Michael Nielsen (1776–1846)

1. Kierkegaards lærere og andre inspiratorer fra ungdomsårene

Kierkegaard's teachers and androther sources of inspiration in his youth

“The judgement on his pedagogical principles is less flattering: he is remembered as ‘Tyrant’ and ‘Despot’.”



"Min gamle Skolemester var en Heros, et Je
vee, vee vee! den Dreng, der ikke kunde sva
paa et ligefrem Spørgsmaal," skriver Kierke
Michael Nielsen, rektor for Borgerdydskole
1844. Som flere af sine kolleger var han pro
og om hans dygtighed som latiner hersker de
Dommen over hans pædagogiske principper
smigrende, han huskes som "Tyran" og "Des
der indtraf tordenvejr, slappedes disciplinen,
Nielsen bange, foldede hænderne og sagde:
saa tier jeg," hvorpå han dog straks tilføjede
taler, saa tier I."

"My old school master was a demigod, an iron man, and woe be unto that boy who could not answer yes or no to a direct question," writes Kierkegaard about Michael Nielsen, the headmaster of Borgerdydskolen [The School of Civic Virtue] from 1813 to 1844. Like several of his colleagues, he was professor in his subject, and there is no doubt of his competence as a Latinist. The judgement on his paedagogical principles is less flattering: he is remembered as "Tyrant" and "Despot". Only when a thunderstorm occurred did discipline relax, because then Nielsen became frightened, folded his hands and said: "When God speaks, I remain silent," whereupon he immediately added: "But when I speak, you remain silent."

“Only when a t did discipline relax, because then N ed, folded his hands and said: ‘When Go lent,’ whereupon he immediately ad k you remain silent.’”

All Of The Subjects Previous Speakers Said I Would Talk About

Technology is the Quest for Control

"To be an Error and to be Cast out is Part of [Nature's] Design."
-*William Blake*

Things Fall Apart -*Chinua Achebe*

EVENTUALLY your desktop computer hardware will ERROR.

EVENTUALLY the information on a hard drive will be CORUPTED.

Amount of useful computer cycles before such an error occurs (CURRENTLY) = large large number

—

Exist robust physical systems which serve as classical computers

Quantum Things Fall Apart Fast

Amount of quantum computer cycles before an error ≠ large number (CURRENTLY?)

Quantum Computing is Bunk

Ways Quantum Computers Fail to Quantum Compute

Decoherence

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\text{environment}\rangle \rightarrow \alpha|0\rangle \otimes |\text{environment}_A\rangle + \beta|1\rangle \otimes |\text{environment}_B\rangle$$

Lack of Unitary Control

attempting to apply unitary evolution **U** instead results in **V**

or (worse) results in non-unitary evolution $\boldsymbol{\rho} \rightarrow \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger$

Measurements are faulty

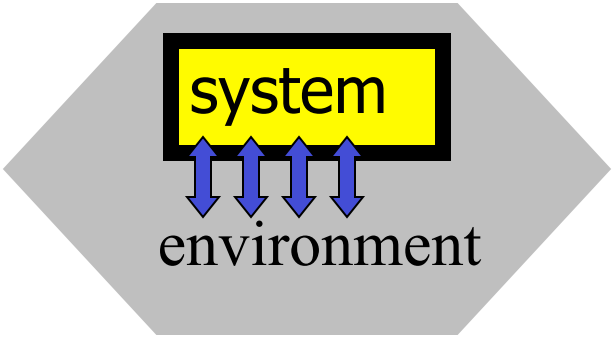
measurement result is noisy, incorrect result obtained

Quantum Computing Disappearing Act

qubits disappear (leakage of computing states)

What is the Problem?

Real quantum systems are open quantum systems!



Full Hilbert space $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$

Evolution is $\mathbf{U} = \exp[-i\mathbf{H}_{SB}t]$

Full Hamiltonian:

System evolution — environment evolution — **Interaction causing DECOHERENCE**

$$\mathbf{H}_{SB} = \mathbf{H}_S \otimes \mathbf{I}_B + \mathbf{I}_S \otimes \mathbf{H}_B + \mathbf{H}_I$$

System's pure states become entangled with environment:

$|\psi_0\rangle = |\psi\rangle_S \otimes |\psi\rangle_B \rightarrow \sum_n c_n |n\rangle \otimes |n\rangle \quad \rho_0 = |\psi_0\rangle_S \, {}_S\langle\psi_0| \rightarrow$ mixed state

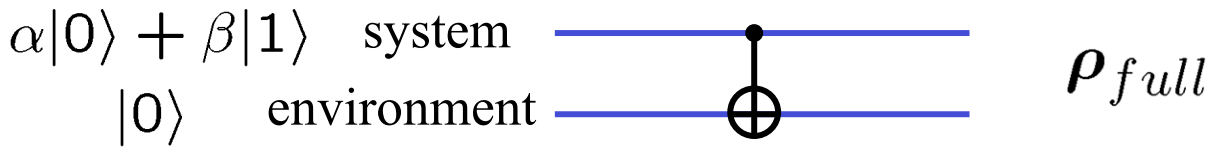
System evolution (by itself) is no longer unitary

Coherent properties of system's quantum information lost

Quantum Error Example

"To err is human, to really foul things up requires a computer." -Anonymous

$$\mathbf{U} = \exp[-i\mathbf{H}_{SB}t]$$



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \rightarrow \alpha|00\rangle + \beta|11\rangle$$

$$\boldsymbol{\rho}_{sys} = \mathrm{Tr}_{env}\left[\boldsymbol{\rho}_{full}\right] = \langle 0_{env}|\boldsymbol{\rho}_{full}|0_{env}\rangle + \langle 1_{env}|\boldsymbol{\rho}_{full}|1_{env}\rangle$$

Before $\mathbf{U}$: $\boldsymbol{\rho}_{sys} = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}$ $\quad \mathbf{U} = \exp[-i\mathbf{H}_{SB}t]$

Not unitary: After $\mathbf{U}$: $\boldsymbol{\rho}_{sys} = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix}$

$$\det\left[\mathbf{U}\boldsymbol{\rho}\mathbf{U}^\dagger\right] = \det\left[\mathbf{U}\right]\det\left[\mathbf{U}^\dagger\right]\det\left[\boldsymbol{\rho}\right] = \det\left[\boldsymbol{\rho}\right]$$

Operator Sum Representation

$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$



Assume initially $\rho = \rho_A \otimes \rho_B$

$$\rho_{out} = \mathrm{Tr}_B[\rho_{AB}] = \mathrm{Tr}_B\left[\mathbf{U}\left(\rho_A \otimes \rho_B\right)\mathbf{U}^\dagger\right]$$

Spectrally decompose ρ_B: $\rho_B = \sum_\nu p_\nu |\nu\rangle\langle\nu|$

$$\begin{aligned}
\rho_{out} &= \sum_\nu p_\nu \mathrm{Tr}_B\left[\mathbf{U}\left(\rho_A \otimes |\nu\rangle\langle\nu|\right)\mathbf{U}^\dagger\right] \\
&= \sum_{\nu,\mu} p_\nu \langle\mu|\mathbf{U}\left(\rho_A \otimes |\nu\rangle\langle\nu|\right)\mathbf{U}^\dagger|\mu\rangle \\
&= \sum_{\nu\mu} \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle \rho_A \langle\nu|\mathbf{U}^\dagger|\mu\rangle\sqrt{p_\nu}
\end{aligned}$$

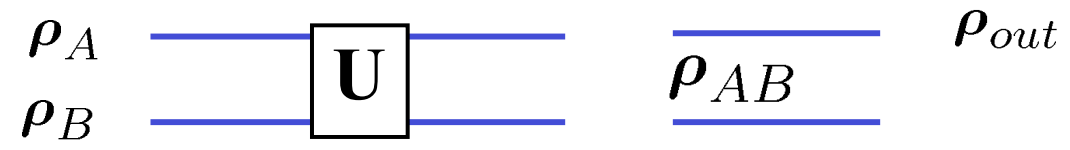
B states

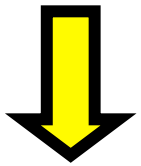
$$\boxed{\rho_{out} = \textstyle\sum_k \mathbf{A}_k \rho_A \mathbf{A}_k^\dagger} \quad \text{where } \mathbf{A}_{k=(\mu,\nu)} = \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle$$

Exercise: Unitarity of $\mathbf{U}$ implies $\boxed{\sum_k \mathbf{A}_k^\dagger \mathbf{A}_k = \mathbf{I}}$

Unmixing Simplification

$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$

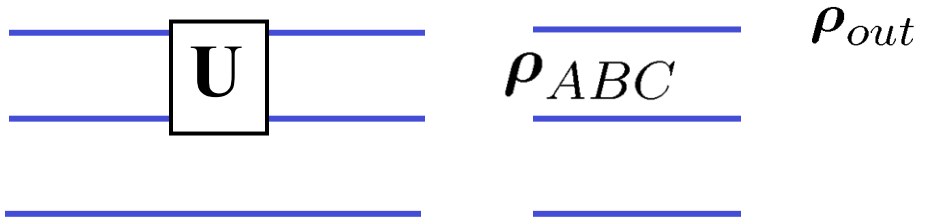




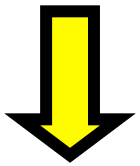
Purification of system B

$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C$

$|\nu\rangle = |\psi_{BC}\rangle$



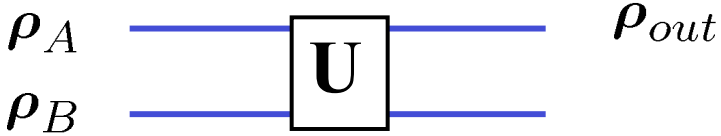
$\rho_{out} = \sum_k \mathbf{A}_k \rho_A \mathbf{A}_k^\dagger$ where $\mathbf{A}_{k=(\mu,\nu)} = \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle$

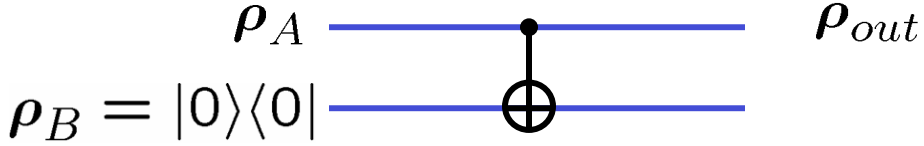


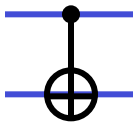
Purification of system B

$\rho_{out} = \sum_k \mathbf{A}_k \rho_A \mathbf{A}_k^\dagger$ where $\mathbf{A}_k = \langle k|\mathbf{U}|\nu\rangle$

OSR Example







$$= |0\rangle\langle 0| \otimes \mathbf{I} + |1\rangle\langle 1| \otimes \mathbf{X} = \frac{1}{2}(\mathbf{II} + \mathbf{ZI} + \mathbf{IX} - \mathbf{ZX})$$

$$\boldsymbol{\rho}_B = \textstyle\sum_\nu p_\nu |\nu\rangle\langle\nu| \rightarrow \boldsymbol{\rho}_B = |0\rangle\langle 0|$$

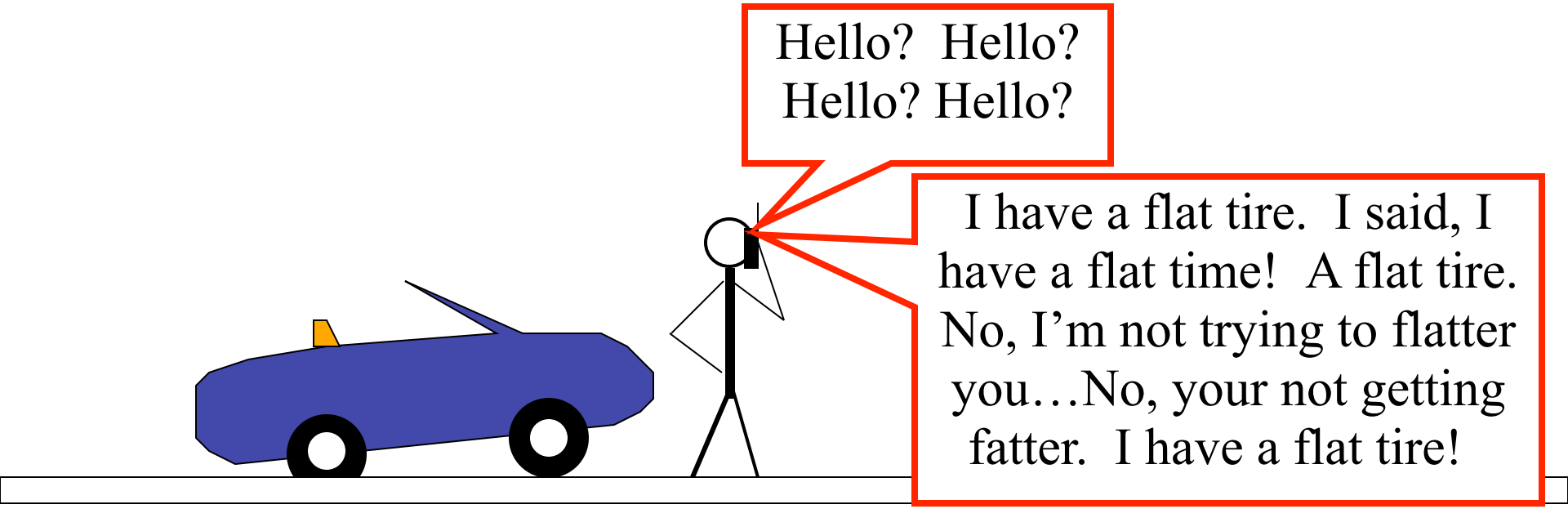
$$\mathbf{A}_{k=(\mu,\nu)} = \sqrt{p_\nu}\langle\mu|\mathbf{U}|\nu\rangle \rightarrow \mathbf{A}_\mu = \langle\mu|\mathbf{U}|0\rangle$$

$$\mathbf{A}_0 = \langle 0|\mathbf{U}|0\rangle = \langle 0|\frac{1}{2}[\mathbf{II} + \mathbf{ZI} + \mathbf{IX} - \mathbf{ZX}]|0\rangle = \frac{1}{2}(\mathbf{I} + \mathbf{Z}) = |0\rangle\langle 0|$$

$$\mathbf{A}_1 = \langle 1|\mathbf{U}|0\rangle = \langle 1|\frac{1}{2}[\mathbf{II} + \mathbf{ZI} + \mathbf{IX} - \mathbf{ZX}]|0\rangle = \frac{1}{2}(\mathbf{I} - \mathbf{Z}) = |1\rangle\langle 1|$$

$$\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_A \mathbf{A}_k^\dagger \rightarrow \boldsymbol{\rho}_{out} = |0\rangle\langle 0|\boldsymbol{\rho}_A|0\rangle\langle 0| + |1\rangle\langle 1|\boldsymbol{\rho}_A|1\rangle\langle 1|$$

Noisy Cell Phone



Communication over a noisy CHANNEL can be overcome via

ENCODING

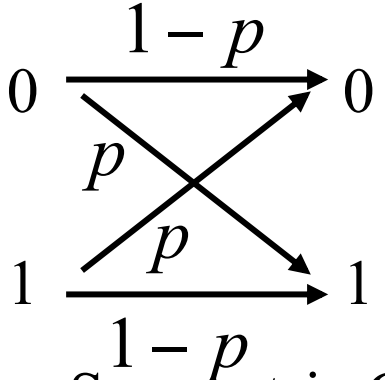
"Hello?" = "Hello? Hello? Hello? Hello?" [using redundancy to encode "Hello"]

Simple Repetition Code

Encode: $0 \rightarrow 000\ldots000$

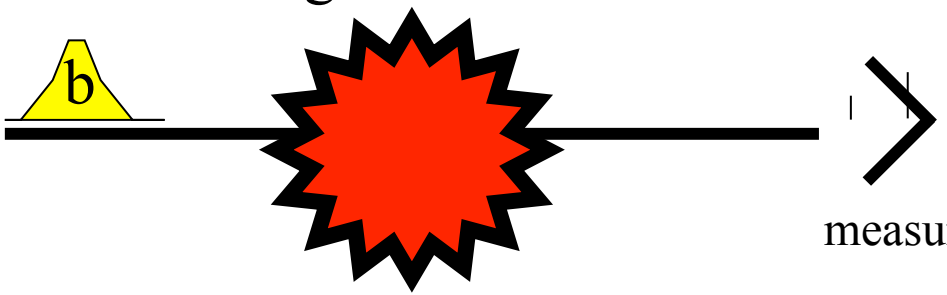
$1 \rightarrow 111\ldots111$

n copies



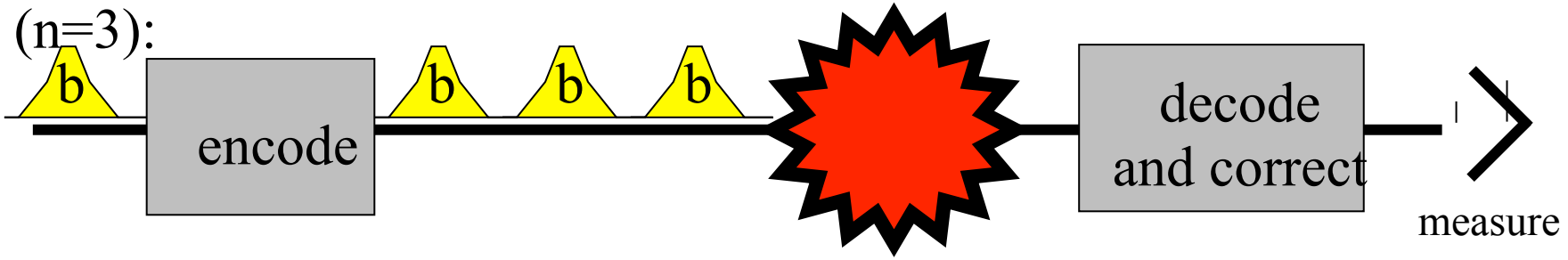
Binary Symmetric Channel

No encoding:



Probability of error = p

Encoding (n=3):



Probability of error

$$= 3p^2(1-p) + p^3 = 3p^2 - 2p^3 < p, p < \tfrac{1}{2}$$

1994 Reasons to be a Pessimist

No cloning:

No cloning of quantum information:

$$\nexists \mathcal{T} || \psi\rangle \rightarrow |\psi\rangle \otimes |\psi\rangle, \forall |\psi\rangle$$

Quantum repetition code impossible:

$$|\psi\rangle \nrightarrow |\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle$$

Continuous set of errors:

Classical errors on digital information are discrete.

Quantum errors on classical information are continuous.

Continuous set of operations:

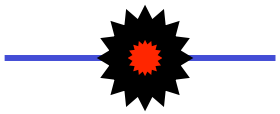
Continuous set of unitary operations: infinite precision needed to implement a gate?

Measurement destroys coherence:

How can one decode without destroying the information?

Baby Steps

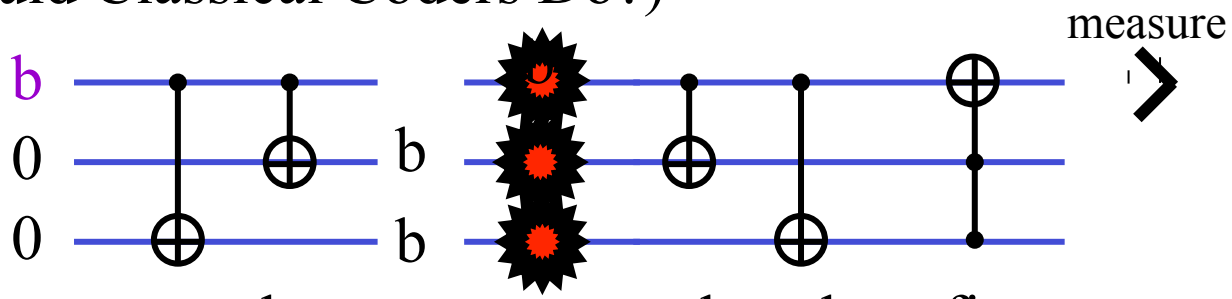
Bit Flip Channel: $\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_{in} \mathbf{A}_k^\dagger = (1-p)\boldsymbol{\rho}_{in} + p\boldsymbol{\sigma}_x \boldsymbol{\rho}_{in} \boldsymbol{\sigma}_x$



$$\mathbf{A}_1 = \sqrt{1-p}\mathbf{I} = \begin{bmatrix} \sqrt{1-p} & 0 \\ 0 & \sqrt{1-p} \end{bmatrix}, \mathbf{A}_2 = \sqrt{p}\boldsymbol{\sigma}_x = \begin{bmatrix} 0 & \sqrt{p} \\ \sqrt{p} & 0 \end{bmatrix}$$

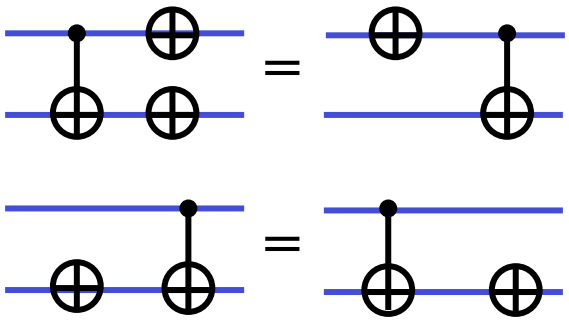
WWCCD? (What Would Classical Coders Do?)

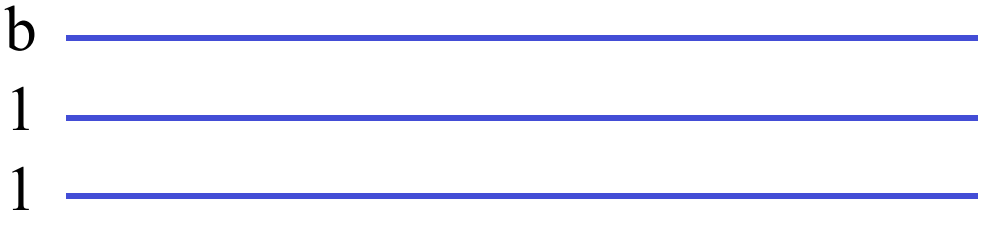
0 □000, 1 □111



100 □ 111 □ 101 □110 □ 110

identities



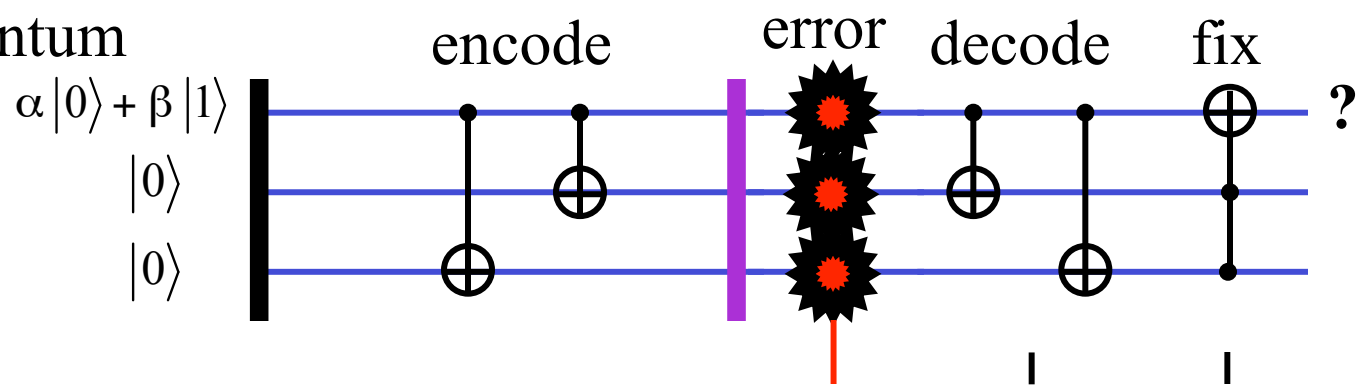


error #@%

Naïve

4. operator identities still hold

Lets be naïve, take classical and move over to quantum



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|0\rangle \otimes |0\rangle \otimes |0\rangle + \beta|1\rangle \otimes |1\rangle \otimes |1\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{XII}{\rightarrow} \alpha|100\rangle + \beta|011\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|11\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IXI}{\rightarrow} \alpha|010\rangle + \beta|101\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IIX}{\rightarrow} \alpha|001\rangle + \beta|110\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle$$

error decode fix

encoded into subspace:

$|0\rangle \rightarrow |000\rangle, |1\rangle \rightarrow |111\rangle$

(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

$\mathbf{XII}|000\rangle = |100\rangle, \mathbf{XII}|111\rangle = |011\rangle$

3. syndrome

OK Wise Guy

What about "phase" errors? $\rho_{out} = (1-p)\rho_{in} + p\sigma_z\rho_{in}\sigma_z$

phase error: $\sigma_z(\alpha|0\rangle + \beta|1\rangle) = (\alpha|0\rangle - \beta|1\rangle)$ …sort of not classical error

Wise guy says "basis change please":

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

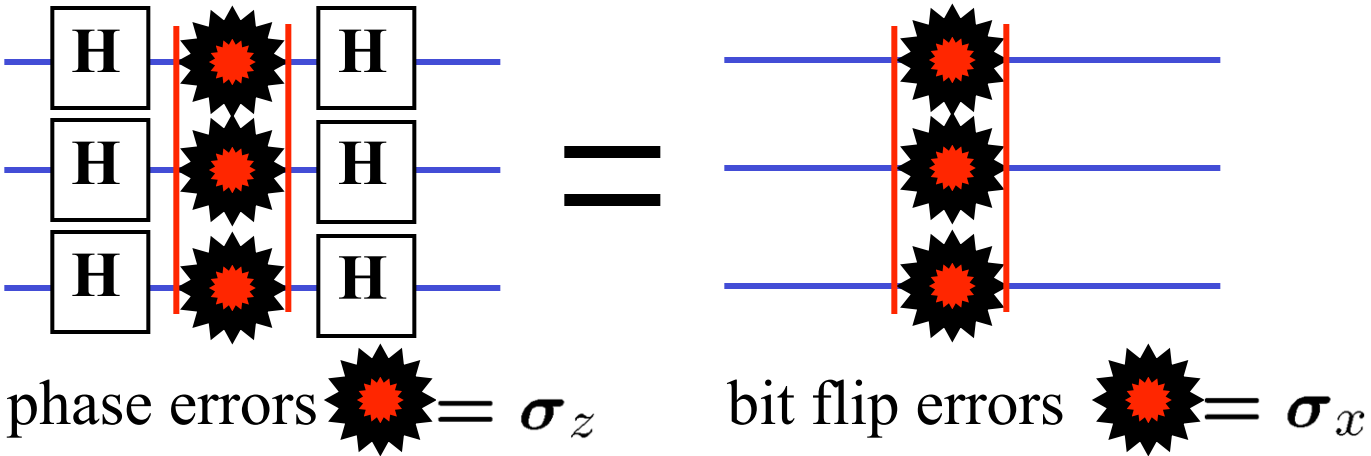
$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\sigma_z|+\rangle = |-\rangle$$

$$\sigma_z|-\rangle = |+\rangle$$

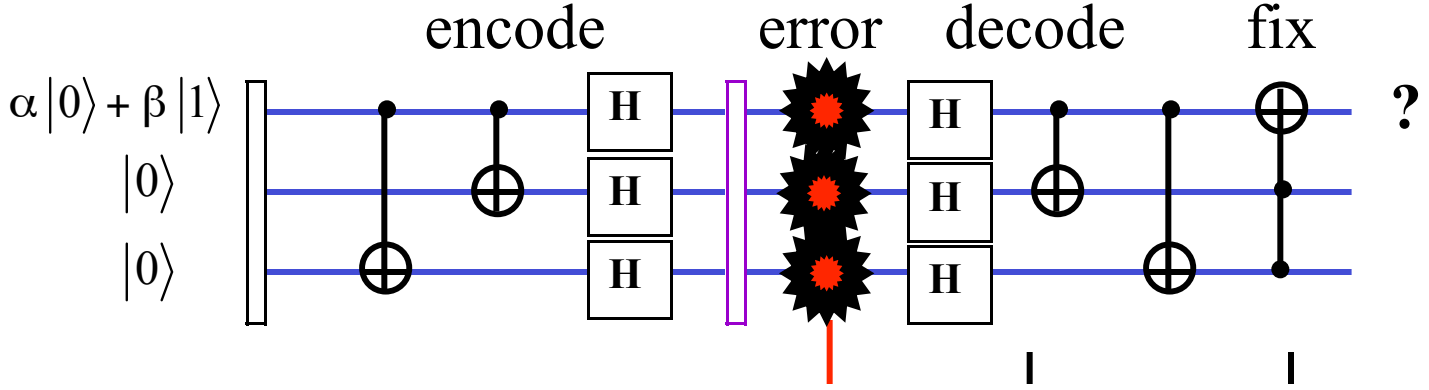
looks like bit flit error in this new basis!

$$\mathbf{H}\sigma_z\mathbf{H} = \sigma_x, \quad \mathbf{H} = |+\rangle\langle 0| + |-\rangle\langle 1|$$



Molly: "I love you, I really love you"

Sam: "Ditto."



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|+\rangle \otimes |+\rangle \otimes |+\rangle + \beta|-\rangle \otimes |-\rangle \otimes |-\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{ZII}{\rightarrow} \alpha|-++\rangle + \beta|+--\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)\,|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|11\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IZI}{\rightarrow} \alpha|+-+\rangle + \beta|-+-\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IIZ}{\rightarrow} \alpha|++-\rangle + \beta|--+\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle$$

error decode fix

<u>encoded into subspace</u>:

$|0\rangle \rightarrow |+++\rangle, |1\rangle \rightarrow |---\rangle$

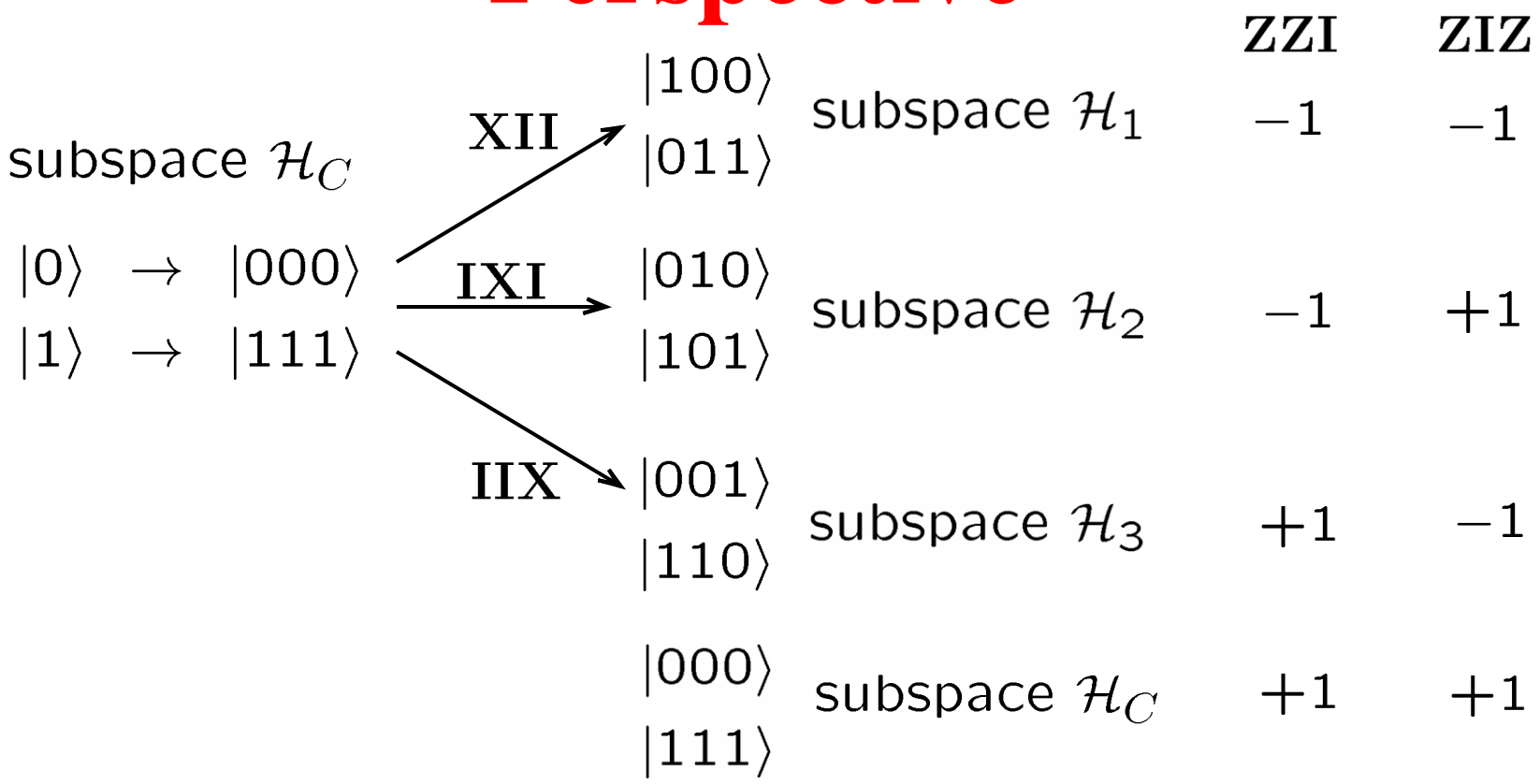
(no-cloning evaded!)

2. <u>errors take to orthogonal subspaces + maintain orthogonality</u>

$\mathbf{ZII}|+++\rangle = |-++\rangle, \mathbf{ZII}|---\rangle = |+--\rangle$

3. <u>syndrome</u>

Perspective



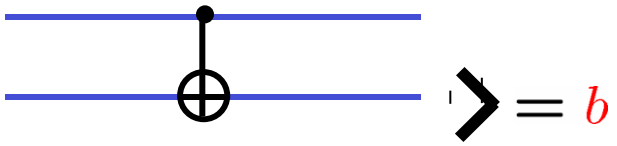
subspaces $\mathcal{H}_C$, $\mathcal{H}_1$, $\mathcal{H}_2$, and $\mathcal{H}_3$ are orthogonal

$\dim \mathcal{H}_C = \dim \mathcal{H}_1 = \dim \mathcal{H}_2 = \dim \mathcal{H}_3 = 2$

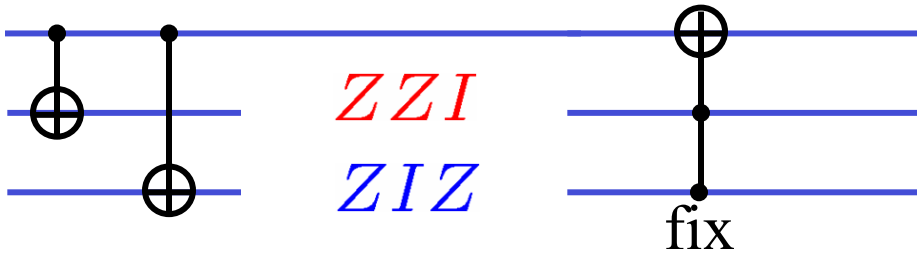
Orthogonal subspaces can be distinguished by measurements

without measuring information encoded into the subspace!!!

Not An Optical Illusion



		b	ZZ
$\|00\rangle$	$\rightarrow$	$\|00\rangle$	$+1$
$\|01\rangle$	$\rightarrow$	$\|01\rangle$	-1
$\|10\rangle$	$\rightarrow$	$\|11\rangle$	-1
$\|11\rangle$	$\rightarrow$	$\|10\rangle$	$+1$



		XII	$\|100\rangle$	$\|111\rangle$	$\|011\rangle$
			$\|011\rangle$	$\|011\rangle$	$\|111\rangle$
$\|0\rangle$	$\rightarrow$ $\|000\rangle$	IXI	$\|010\rangle$	$\|010\rangle$	$\|010\rangle$
$\|1\rangle$	$\rightarrow$ $\|111\rangle$		$\|101\rangle$	$\|110\rangle$	$\|110\rangle$
		IIX	$\|001\rangle$	$\|001\rangle$	$\|001\rangle$
		error	$\|110\rangle$	$\|101\rangle$	$\|101\rangle$

Encoding Away Your Ills

3 qubit bit flip code	3 qubit phase flip code
$\|0\rangle \rightarrow \|000\rangle$	$\|0\rangle \rightarrow \|+++\rangle$
$\|1\rangle \rightarrow \|111\rangle$	$\|1\rangle \rightarrow \|---\rangle$

phase errors **ZII**, **IZI**, **IIZ** act as ■ on bit flip code qubits:

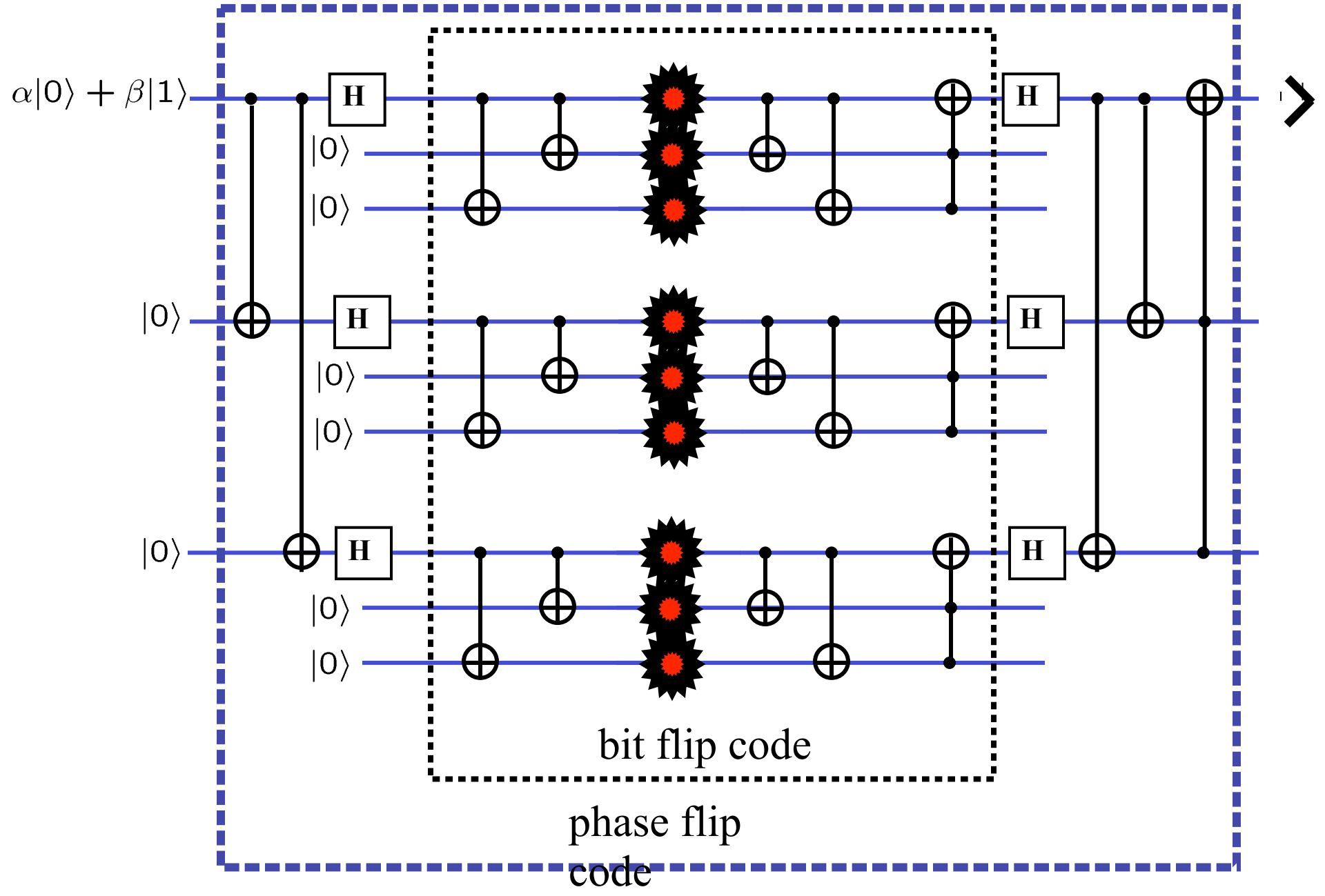
$$|0\rangle_B = |000\rangle \qquad \mathbf{ZII}|0\rangle_B = |0\rangle_B$$
$$|1\rangle_B = |111\rangle \qquad \mathbf{ZII}|1\rangle_B = -|1\rangle_B$$

define:
$$|p\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B + |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$
$$|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B - |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$$

Shor Code: (Peter Shor, 1995)

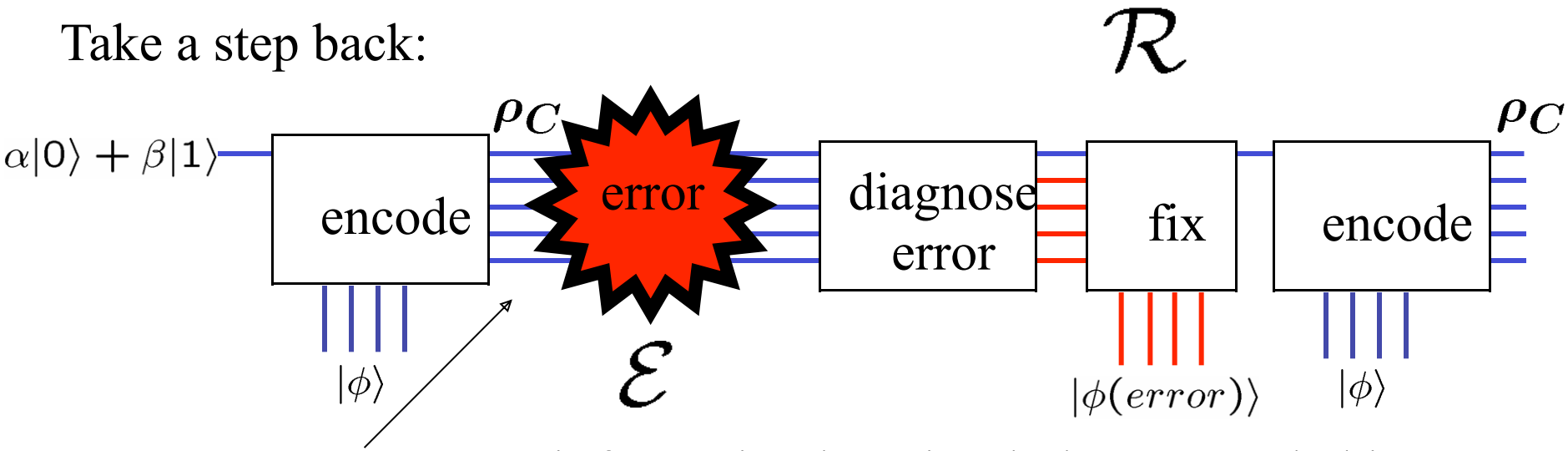
$$|0\rangle \rightarrow |ppp\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$
$$|1\rangle \rightarrow |mmm\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

Inside Shor


$\alpha|0\rangle + \beta|1\rangle$
$|0\rangle$
$|0\rangle$
H
H
H
H
H
H
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
bit flip code
phase flip code

Do the Hokey Pokey

Take a step back:



Here quantum information has simply been encoded into ρ_C

Then an error occurs

$$\mathcal{E}[\rho_C] = \sum_k \mathbf{E}_k \rho_C \mathbf{E}_k^\dagger$$

Then a recovery procedure

$$\mathcal{R}[\mathcal{E}[\rho_C]] = \sum_l \mathbf{R}_l \sum_k \mathbf{E}_k \rho_C \mathbf{E}_k^\dagger \mathbf{R}_l^\dagger$$

An miraculously we recover the encoded state ρ_C

$$\mathcal{R}[\mathcal{E}[\rho_C]] = \rho_C$$

Lets Get Definitional

A quantum code C is simply a ~~subspace~~ subsystem

Quantum Error Correcting Code

A quantum error correcting code C for the quantum operation $\mathcal{E}$ is a code for which there exists a recovery operator $\mathcal{R}$ such that $\mathcal{R} \circ \mathcal{E}$ acts proportional to identity on the codespace.

Iff Condition for Quantum Error Correcting Code

A code C with orthonormal basis $\{|i\rangle\}$ is a quantum error correcting code for the quantum operation $\mathcal{E}$ with operation elements $\{\mathbf{E}_k\}$ iff

$$\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = C_{ab}\delta_{ij}$$

Necessary

$\mathcal{R}[\mathcal{E}[\boldsymbol{\rho}_C]] \propto \boldsymbol{\rho}_C$ for $\boldsymbol{\rho}_C$ with support over the code C:

$$\sum_{kl} \mathbf{R}_l \mathbf{E}_k \boldsymbol{\rho}_c \mathbf{E}_k^\dagger \mathbf{R}_l^\dagger = c\boldsymbol{\rho}_c$$

⇓

$\sum_{kl} \mathbf{R}_l \mathbf{E}_k \mathbf{P}_C \boldsymbol{\rho} \mathbf{P}_C \mathbf{E}_k^\dagger \mathbf{R}_l^\dagger = c\mathbf{P}_C \boldsymbol{\rho} \mathbf{P}_C$ where $\mathbf{P}_C = \sum_{i \in C} |i\rangle\langle i|$ is projector onto code C.

⇓

This must hold for all $\boldsymbol{\rho}$, so $\mathbf{R}_l \mathbf{E}_k \mathbf{P}_C = \lambda_{lk} \mathbf{P}_C$

⇓

$$\mathbf{P}_C \mathbf{E}_m^\dagger \mathbf{R}_l^\dagger \mathbf{R}_l \mathbf{E}_k \mathbf{P}_C = \lambda_{lm}^* \lambda_{lk} \mathbf{P}_C$$

⇓

Use $\sum_l \mathbf{R}_l^\dagger \mathbf{R}_l = \mathbf{I}$, to obtain $\mathbf{P}_C \mathbf{E}_m^\dagger \mathbf{E}_k \mathbf{P}_C = \sum_l \lambda_{lm}^* \lambda_{lk} \mathbf{P}_C$

⇓

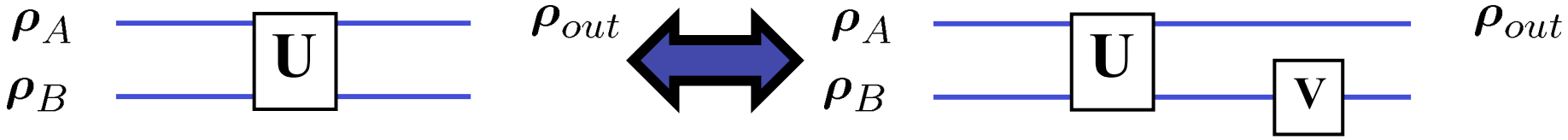
$$\langle i|\mathbf{E}_m^\dagger \mathbf{E}_k|j\rangle = C_{mk}\delta_{ij}$$

OSR Unitary Freedom

OSR: $\mathbf{A}_k = \langle k|\mathbf{U}|\nu\rangle$ $\quad$ $\boldsymbol{\rho}_{out} = \sum_k \mathbf{A}_k \boldsymbol{\rho}_{in} \mathbf{A}_k^\dagger$

Whose $|k\rangle$?

$|k\rangle$ was from Trace operation: basis arbitrary.



Equivalent from system A's perspective

$$|l\rangle = \sum_k V_{kl}|k\rangle, |k\rangle = \sum_l V_{kl}^*|l\rangle, \quad VV^\dagger = I$$

New OSR: $\mathbf{B}_l = \langle l|\mathbf{U}|\nu\rangle$

$$\mathbf{A}_k = \langle k|\mathbf{U}|\nu\rangle = \sum_l V_{kl}\langle l|\mathbf{U}|\nu\rangle = \sum_l V_{kl}\mathbf{B}_l$$

Two OSR's $\{\mathbf{A}_k\}$ and $\{\mathbf{B}_k\}$ are equivalent iff there exists a unitary matrix V_{kl} such that $\mathbf{A}_k = \sum_l V_{kl}\mathbf{B}_l$.

Diagonal Error Condition

Error correcting code criteria: $\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = C_{ab}\delta_{ij}$

$$C_{ab}\delta_{ij} = \langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = \left(\langle j|\mathbf{E}_b^\dagger \mathbf{E}_a|i\rangle\right)^* = C_{ba}^*\delta_{ij}$$

$C_{ab} = C_{ba}^*$: C is Hermitian

Change of OSR basis: $\mathbf{F}_k = \sum_k u_{ka}\mathbf{E}_a$

Error correcting code criteria: $\langle i|\mathbf{F}_k^\dagger \mathbf{F}_l|j\rangle = D_{kl}\delta_{ij}$

$$\langle i|\mathbf{F}_k^\dagger \mathbf{F}_l|j\rangle = \sum_{ab} u_{ka}^* u_{lb}\langle i|\mathbf{E}_a^\dagger \mathbf{E}_b|j\rangle = \sum_{ab} u_{ka}^* C_{ab} u_{bl}\delta_{ij}$$

$$\sum_{ab} u_{ka}^* C_{ab} u_{bl} = D_{kl} = d_k\delta_{kl}$$

$$\boxed{\langle i|\mathbf{F}_a^\dagger \mathbf{F}_b|j\rangle = d_a\delta_{ab}\delta_{ij}}$$

Sufficient

Projector onto the code: $\mathbf{P}_C = \sum_{i \in C} |i\rangle\langle i|$

$$\langle i|\mathbf{F}_a^\dagger \mathbf{F}_b|j\rangle = d_a \delta_{ab} \delta_{ij}$$

Projector onto the code followed by error $\mathbf{F}_a$:

$$\mathbf{F}_a \mathbf{P}_C = \mathbf{U}_a \sqrt{\mathbf{P}_C \mathbf{F}_a^\dagger \mathbf{F}_a \mathbf{P}_C} = \mathbf{U}_a \sqrt{d_a} \mathbf{P}_C$$

Polar decomposition: $\mathbf{A} = \mathbf{U}\sqrt{\mathbf{A}^\dagger \mathbf{A}}$, $\mathbf{U}$ unitary

Define the projectors $\mathbf{P}_a = \mathbf{U}_a \mathbf{P}_C \mathbf{U}_a^\dagger$ which are orthogonal:

$$\mathbf{P}_a \mathbf{P}_b = \mathbf{U}_a \mathbf{P}_C \mathbf{U}_a^\dagger \mathbf{U}_b \mathbf{P}_C \mathbf{U}_b^\dagger = \frac{1}{\sqrt{d_a d_b}} \mathbf{U}_a \mathbf{P}_C \mathbf{F}_a^\dagger \mathbf{F}_b \mathbf{P}_C \mathbf{U}_b^\dagger = \delta_{ab}$$

$$\mathbf{U}_a^\dagger \mathbf{P}_a \mathbf{F}_a \mathbf{P}_C = \sqrt{d_a} \mathbf{P}_C$$

fix/diagnose error/error/code

$$\mathbf{U}_b^\dagger \mathbf{P}_b \mathbf{F}_a \mathbf{P}_C = 0$$

fix/diagnose no error/error/code

$$\mathbf{U}_a^\dagger \mathbf{P}_a \mathbf{F}_a \mathbf{P}_C = \sqrt{d_a}\mathbf{P}_C \qquad \mathbf{U}_b^\dagger \mathbf{P}_b \mathbf{F}_a \mathbf{P}_C = 0$$

Recover $\mathcal{R}$ consists of measurement $\mathcal{P}$ followed by unitary fixing conditional on the measurement outcome:

Padding may be needed.

$$\mathcal{P}[\boldsymbol{\rho}] = \sum_a \mathbf{P}_a \boldsymbol{\rho} \mathbf{P}_a^\dagger$$

$$\mathcal{R}[\boldsymbol{\rho}] = \sum_a \mathbf{U}_a^\dagger \mathbf{P}_a \boldsymbol{\rho} \mathbf{P}_a \mathbf{U}_a$$

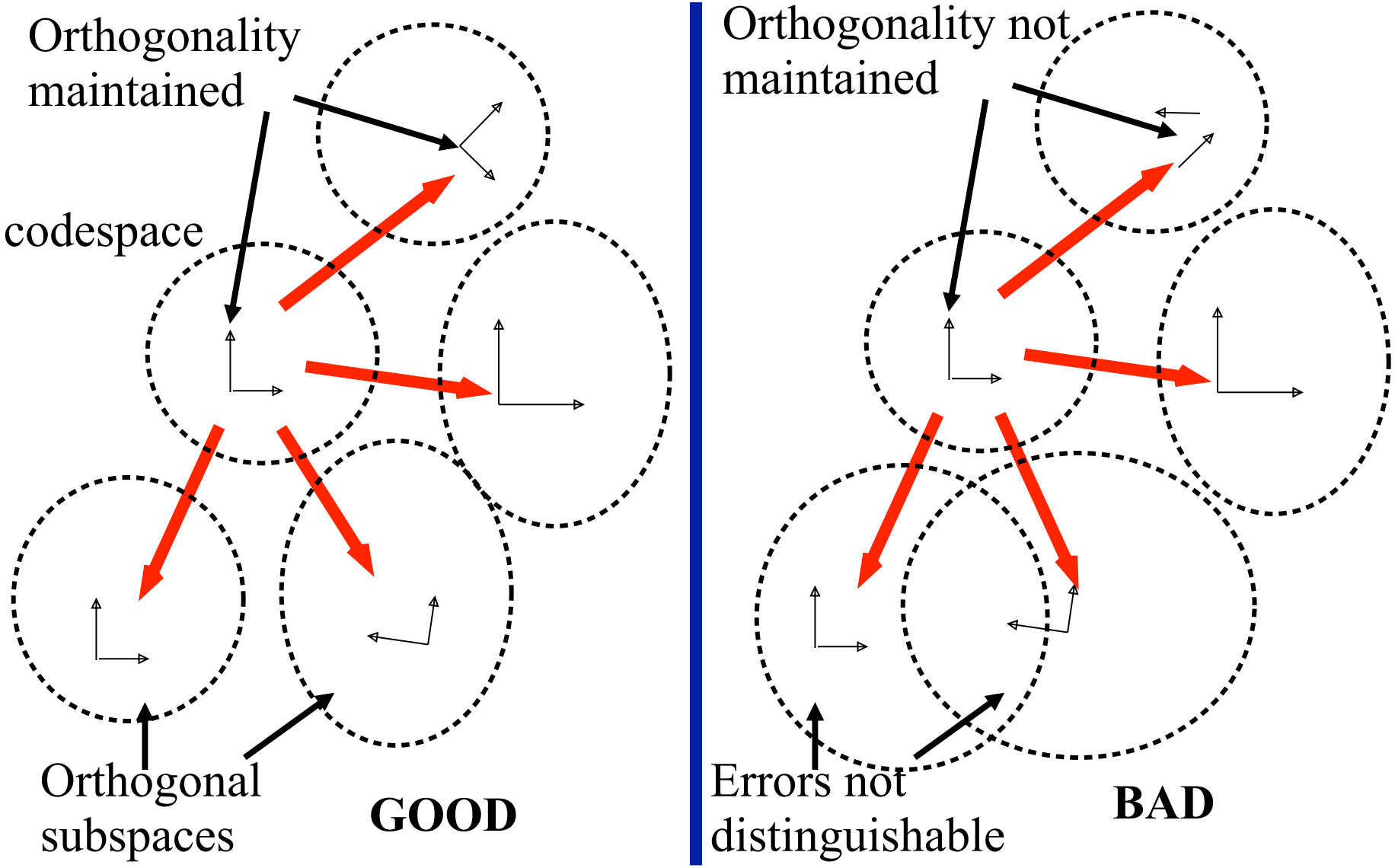
$$\mathcal{R} \circ \mathcal{F}[\boldsymbol{\rho}_C] = \textstyle\sum_{ab} \mathbf{U}_a^\dagger \mathbf{P}_a \mathbf{F}_b \mathbf{P}_C \boldsymbol{\rho}_C \mathbf{P}_C \mathbf{F}_b^\dagger \mathbf{P}_a \mathbf{U}_a \propto \boldsymbol{\rho}_C$$

recovery error

Explicit recovery operator derived.

Pictures

$$\langle i|\mathbf{F}_a^\dagger \mathbf{F}_b|j\rangle = D_a \delta_{ab}\delta_{ij}$$



Digitizing Errors

So far we have designed our error correction for a specific OSR evolution

$$\boldsymbol{\rho} \to \sum_a \mathbf{F}_a \boldsymbol{\rho} \mathbf{F}_a^\dagger$$

such that for our recover operator $\mathcal{R}$

$$\mathcal{R}\left[\sum_a \mathbf{F}_a \boldsymbol{\rho}_C \mathbf{F}_a^\dagger\right] \propto \boldsymbol{\rho}_C$$

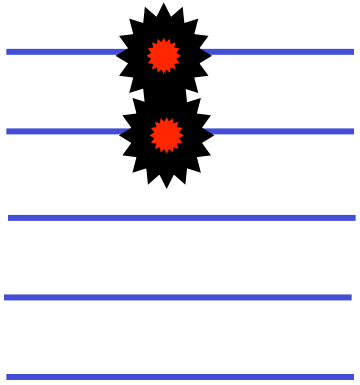
$$\mathbf{U}_b^\dagger \mathbf{P}_b \mathbf{F}_a \mathbf{P}_C = \delta_{ab}\sqrt{d_a}\mathbf{P}_C$$

If we have another OSR $\rho \to \sum_c \mathbf{E}_c \boldsymbol{\rho} \mathbf{E}_c^\dagger$ whose elements are $\mathbf{E}_c = \sum_a v_{ca}\mathbf{F}_a$, then

$$\mathbf{U}_b^\dagger \mathbf{P}_b \mathbf{E}_c \mathbf{P}_C = \mathbf{U}_b^\dagger \mathbf{P}_b \sum_a v_{ca}\mathbf{F}_a \mathbf{P}_C = \delta_{ab} v_{ca}\sqrt{d_a}\mathbf{P}_C$$

code corrects OSR operators $\{\mathbf{E}_a\} \to$ code corrects OSR operators $\{\sum_b v_{ba}\mathbf{E}_a\}$.

How Have We Helped?



Suppose error acts on one of many qubits

$$\boldsymbol{\rho} \to \sum_k \mathbf{E}_k \boldsymbol{\rho} \mathbf{E}_k^\dagger, \quad \mathbf{E}_k = \mathbf{E}_k \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}$$

Single qubit error correcting code:

$$\text{If } \mathbf{E}_k = e_0 \mathbf{I} + \textstyle\sum_\alpha e_\alpha \boldsymbol{\sigma}_\alpha, \text{ then } \mathcal{R}\left[\sum_k \mathbf{E}_k \boldsymbol{\rho} \mathbf{E}_k^\dagger\right] = \boldsymbol{\rho}$$

$$\mathcal{R}[\boldsymbol{\rho}] = \sum_a \mathbf{U}_a^\dagger \mathbf{P}_a \boldsymbol{\rho} \mathbf{P}_a \mathbf{U}_a$$

Notion of "no error outcome" for a single qubit error, i.e. the measurement projector $\mathbf{P}_0 = \mathbf{P}_C$, implies "probability of no error" $= 1 - p$.

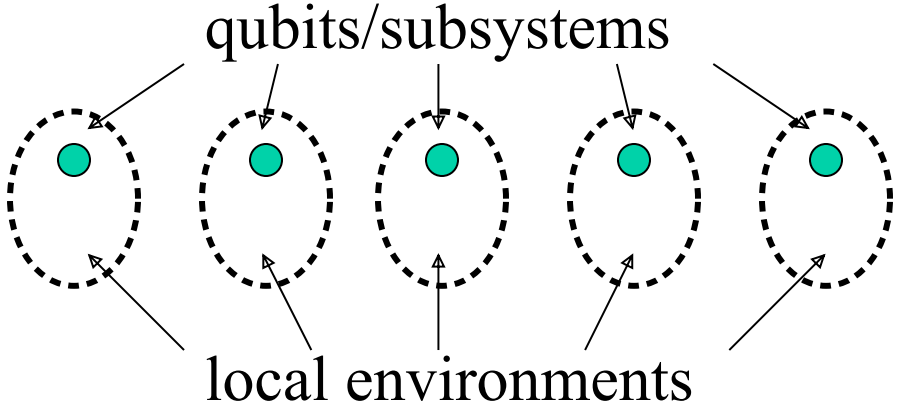
Now suppose two errors act "independently":

$$\boldsymbol{\rho} \to \sum_l \mathbf{F}_l \sum_k \mathbf{E}_k \boldsymbol{\rho} \mathbf{E}_k^\dagger \mathbf{F}_l^\dagger, \quad \mathbf{E}_k = \mathbf{E}_k \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}, \mathbf{F}_l = \mathbf{I} \otimes \mathbf{F}_l \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}$$

Notion of no two error probability $= (1-p)^2$

Independent Error Model

WARNING: "An approximation only a physicist could love."



Evolution dominated by interaction with different environment modes which do not interact with each other:

$$\mathcal{E}[\boldsymbol{\rho}] = \sum_{k_1,k_2,\cdots,k_n} \mathbf{E}_{k_1} \otimes \mathbf{E}_{k_2} \otimes \cdots \otimes \mathbf{E}_{k_n} \boldsymbol{\rho} \mathbf{E}_{k_1}^{\dagger} \otimes \mathbf{E}_{k_2}^{\dagger} \otimes \cdots \otimes \mathbf{E}_{k_n}^{\dagger}$$

Notion of "no error probability" gives credence to standard argument for error correction success.

Really we can weaken this requirement (exponentially decreasing correlation [time and space] between errors).

Coming Attractions

How to become a quantum error correction coder in minutes (get rich quick scheme, I think not)

the Pauli stabilizer formalism

Doing something useful with a quantum codes

the normalizer

Finding out that the something which was useful isn't

the Knill-Gottesman theorem

Doing what many thought impossible

intro to fault-tolerance